

"Diagrammatic Dialog Design Notations & Computer Mediated Communications"

Based on the slide titled "**Diagrammatic Dialog Design Notations & Computer Mediated Communications**", here is a complete and organized list of **short and descriptive questions with their answers** covering all topics from the slide.

✔ Short Questions with Answers

1. What is dialog in HCI?

Answer:

Dialog refers to the **interactive communication between a user and a computer system**, including what the system does (semantics) and how it looks (presentation).

2. What are dialog design notations?

Answer:

Dialog design notations are methods used to **describe and model system-user interactions**, often in diagrammatic or textual form.

3. What are the two types of dialog notations?

Answer:

1. **Diagrammatic Notations** (e.g., State Charts, Petri Nets)
 2. **Textual Notations** (e.g., Grammars, Production Rules)
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4. What are the three levels of dialog in HCI?

Answer:

1. **Lexical level** – Symbols or characters used.
 2. **Syntactic level** – Structure/order of those symbols.
 3. **Semantic level** – Meaning behind the input/output.
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5. What is a State Transition Network (STN)?

Answer:

A diagram showing **states as circles** and **transitions/events as arrows** to model system or user behavior across multiple states.

6. What is a Petri Net?

Answer:

A Petri Net is a **graph-based model** that represents states (places), transitions (events), and tokens (current state markers), useful for modeling **concurrent systems**.

7. What are the key elements of Petri Nets?

Answer:

- **Places:** Circles
 - **Transitions:** Rectangles
 - **Tokens:** Dots in places
 - **Arcs:** Connectors between places and transitions
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8. What is a StateChart?

Answer:

A **hierarchical state machine diagram** that models **complex, concurrent, and reactive systems**, extending basic finite state machines.

9. What are Flowcharts used for?

Answer:

Flowcharts are **step-by-step diagrams** that represent processes, algorithms, or task flows in a sequential manner.

10. What is Jackson Structured Design (JSD)?

Answer:

A **structured diagramming method** used for system and dialog design, especially for modeling **data-driven systems and user interactions**.

11. What is asynchronous communication in CMC?

Answer:

Communication that does **not happen in real time** (e.g., emails, downloading files).

12. What is synchronous communication in CMC?

Answer:

Communication that happens **in real time** (e.g., chat, voice calls, video calls).

Descriptive Questions with Answers

1. What is the importance of dialog design notations in HCI?

Answer:

Dialog design notations help in:

- Analyzing user-system interactions
 - Detecting inconsistent actions or errors
 - Structuring the behavior and flow of user interfaces
 - Enhancing usability and system clarity
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2. Describe the components of Petri Nets with an example.

Answer:

Components:

- **Places (P):** Represent system states (e.g., "coin inserted")
- **Transitions (T):** Events that move tokens from one place to another
- **Tokens:** Represent the current state/condition
- **Arcs:** Direct connections between places and transitions

Example: Coffee Machine

- P1: Coin Inserted → P2: Coffee Selected → P3: Brewing → P4: Dispensed
 - T1–T4 are the transitions triggering each step
 - Tokens move as each action completes
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3. Explain the benefits of StateCharts over basic state machines.

Answer:

- **Handle concurrency** via parallel states
- **Hierarchical modeling** using superstates and substates

- **Improved clarity and modularity** for large, complex systems
- **Easier maintenance and updates**

4. Compare Synchronous and Asynchronous Communication in CMC.

Aspect	Synchronous	Asynchronous
Timing	Real-time	Delayed
Examples	Video calls, chat	Email, downloading, forum posts
Feedback	Instant	Delayed
Use case	Urgent or collaborative tasks	Tasks with time flexibility

5. What are the applications of online communication tools?

Answer:

- **Email** – Formal and asynchronous communication
- **Messaging Systems** – Instant chatting (WhatsApp, Messenger)
- **Video Conferencing** – Virtual meetings and interviews (Zoom, Teams)
- **Collaborative Tools** – Shared editing and task tracking (Google Docs, Trello)

6. What are the advantages of using Petri Nets in system design?

Answer:

- Visually clear and structured
- Helps in modeling **concurrency** and **parallel processes**
- Can simulate and detect **deadlocks or inefficiencies**
- Easily maps to real-world task workflows

Let me know if you'd like this compiled into a printable study guide, or need any diagrams or summaries added! 😊📖